

EE 102 LAB

Experiment - 6 Operational Amplifier Circuits

Group 13

ALA - B

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OBJECTIVE :-

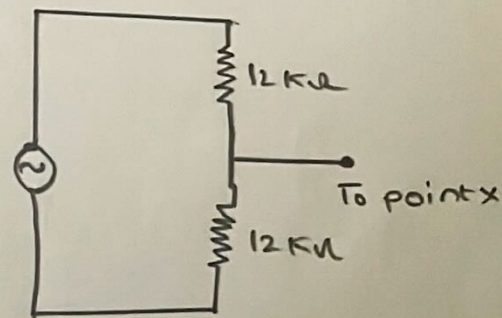
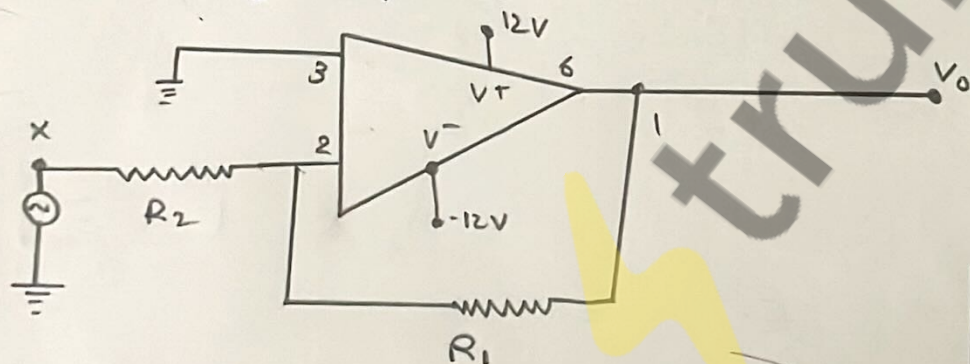
- To study inverting and non-inverting amplifier circuits with Op-amp.
- To study a non-inverting high pass Filter circuit made using op-amp.

MATERIALS :-

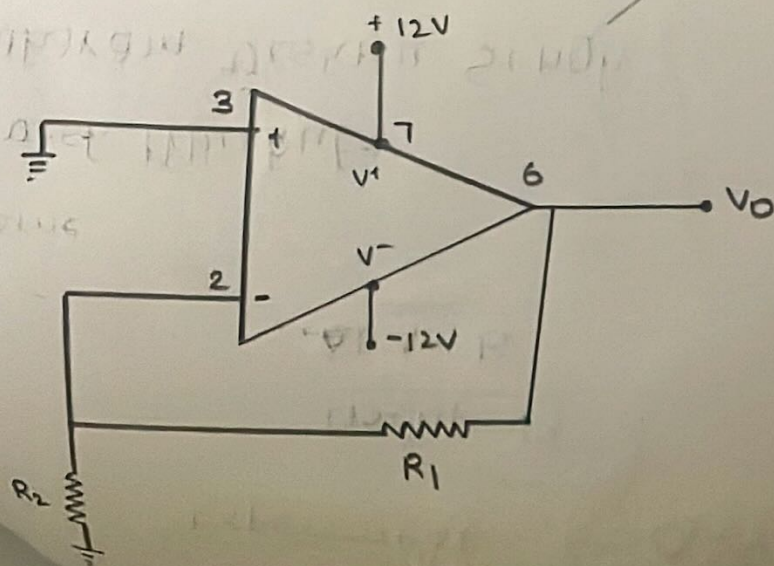
- Breadboard, Function generator, Oscilloscope, DC supply
- LM741 (Op-amp), $1K\Omega$, $10K\Omega$, $12K\Omega$ (2), $100K\Omega$, $0.1\mu F$

CIRCUIT DIAGRAMS :-

For: Inverting Amplifier



For: non-inverting Amplifier



Ans 1.

Considering ideal op-Amp.

$$V_+ = V_- = 0$$

from fig., $V_x = \frac{V_i}{2}$

Applying KVL

$$\frac{V_o - V_+}{R_1} = \frac{V_- - V_x}{R_2}$$

$$\Rightarrow \frac{V_o}{R_1} = \frac{-V_i}{2R_2}$$

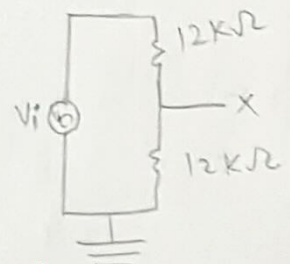
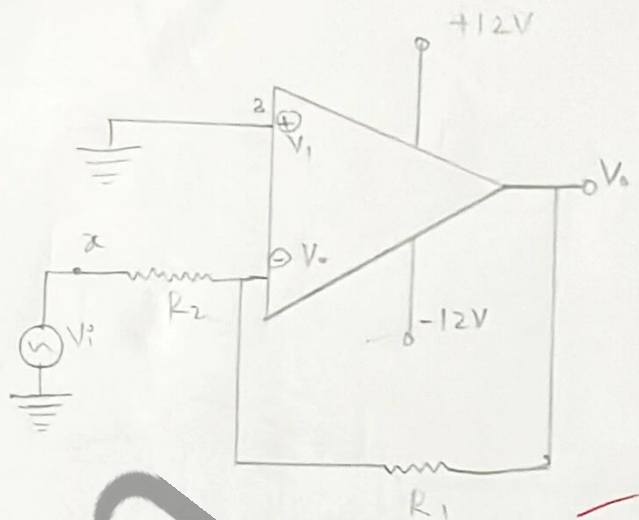
$$\Rightarrow \boxed{\frac{V_o}{V_i} = -\frac{R_1}{2R_2}}$$

(i) $R_1 = 100K\Omega$ & $R_2 = 10K\Omega$

$$\therefore \frac{V_o}{V_i} = \frac{-100}{2 \times 10} = -5$$

(ii) $R_1 = 10K\Omega$ & $R_2 = 1K\Omega$

$$\therefore \frac{V_o}{V_i} = \frac{-10}{2 \times 1} = -5$$



$R_{th} = 6$

$V_{th} = \frac{V_i}{2} = V_x$

② considering ideal op-Amp

$V_+ = V_- = V_x = \frac{V_i}{2}$ (from fig.)

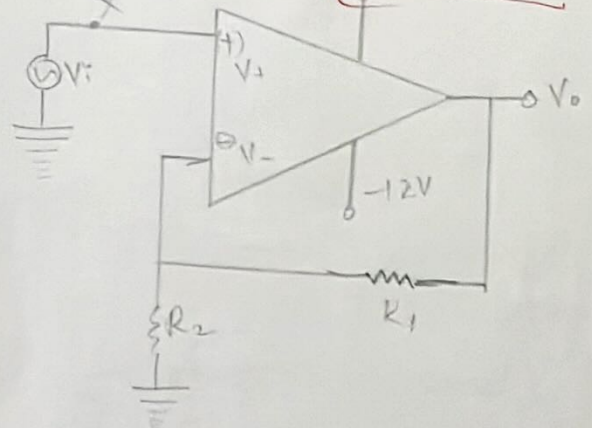
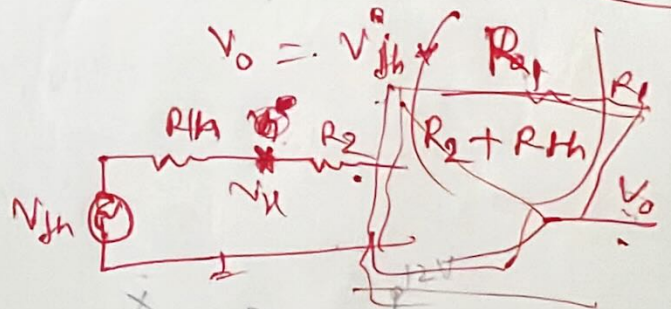
Applying KVL

$$\frac{V_o - V_-}{R_1} = \frac{V_+ - 0}{R_2}$$

$$\Rightarrow \frac{V_o - \frac{V_i}{2}}{R_1} = \frac{\frac{V_i}{2}}{R_2}$$

$$\Rightarrow R_2 V_o = \left(\frac{R_1 + R_2}{2} \right) V_i$$

$$\Rightarrow \boxed{\frac{V_o}{V_i} = \frac{R_2 + R_1}{2R_2}}$$



$$V_o = \left(\frac{100}{10+6} \right) \times \frac{0.2}{2}$$

$$= \frac{100}{16} \times 0.1 = 0.625$$

RT-A: Inverting Amplifier

$$R_1 = 10\text{K}\Omega \quad R_2 = 1\text{K}\Omega \quad V_i = 200\text{mV (V}_{p-p}) \quad \text{frequency} = 1\text{KHz}$$

Graph drawn, $V_o(p-p) = 2\text{V}$

$$A = V_o/V_i = \frac{-1\text{V}}{200\text{mV}} = -5 \rightarrow \text{Voltage amplification (A}_v\text{)} = -5$$

$$\text{For } V_i = 100\text{mV (V}_{p-p}) \rightarrow V_o = 0.44\text{V (p-p)}$$

$$A = \frac{V_o}{V_i} = \frac{-0.44\text{V}}{50\text{mV}} = -8.8 \rightarrow \text{Voltage Amplification} \rightarrow A_v = -8.8$$

$$\text{For } V_i = 300\text{mV (V}_{p-p}) \rightarrow V_o = 3\text{V (p-p)}$$

$$A = \frac{V_o}{V_i} = \frac{-1.5\text{V}}{150\text{mV}} = -10 \rightarrow \text{Voltage Amplification (A}_v\text{)} = -10$$

② $R_1 = 100\text{K}\Omega \quad R_2 = 10\text{K}\Omega \quad V_i = 300\text{mV (p-p)} \quad \text{frequency} = 1\text{KHz}$

$$V_o(p-p) = 3\text{V}$$

$$A = V_o/V_i = \frac{-1.5\text{V}}{150\text{mV}} = -10 \rightarrow \text{Voltage amplification (A}_v\text{)} = -10$$

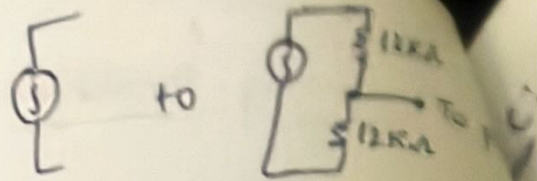
$$\text{for } V_i = 200\text{mV (p-p)} \quad V_o = 2\text{V (p-p)}$$

$$A = \frac{V_o}{V_i} = \frac{-1\text{V}}{100\text{mV}} = -10 \rightarrow \text{Voltage amplification (A}_v\text{)} = -10$$

$$\text{for } V_i = 100\text{mV (p-p)} \quad V_o = 0.9\text{V (p-p)}$$

$$A = \frac{V_o}{V_i} = \frac{-0.45\text{V}}{50\text{mV}} = -9 \rightarrow \text{Voltage amplification (A}_v\text{)} = -9$$

③ Now changing V_i circuit from



i) $R_1 = 100K\Omega$ $R_2 = 10K\Omega$ $V_i = 200mV(p-p)$ frequency = $1KHz$

$$V_o = 0.56V(p-p)$$

$$A = \frac{V_o}{V_i} = \frac{-0.28V}{100mV} = -2.8 \rightarrow \text{Voltage Amplification } (A_v) = -2.8$$

ii) $R_1 = 10K\Omega$ $R_2 = 1K\Omega$ $V_i = 200mV(p-p)$ frequency = $1KHz$

$$V_o = 0.14V(p-p)$$

$$A = \frac{V_o}{V_i} = \frac{0.07}{100mV} = -0.7 \rightarrow \text{Voltage amplification } (A_v) = -0.7$$

PART B: Non-Inverting Amplifier

① $R_1 = 10K\Omega$ $R_2 = 1K\Omega$ $V_i = 200mV(p-p)$ frequency = $1KHz$

$$V_o = 2.1V(p-p)$$

$$A = \frac{V_o}{V_i} = \frac{1.05V}{100mV} = 10.5 \rightarrow \text{Voltage amplification } (A_v) = 10.5$$

For $V_i = 100mV(p-p) \rightarrow V_o = 2V(p-p)$

$$A = \frac{V_o}{V_i} = \frac{2V}{100mV} = 10 \rightarrow \text{Voltage amplification } (A_v) = 10$$

For $V_i = 300mV(p-p) \rightarrow V_o = 3.4(p-p)$

$$A = \frac{V_o}{V_i} = \frac{3.4V}{300mV} = 11.33 \rightarrow \text{Voltage amplification } (A_v) = 11.33$$

② $R_1 = 100\text{K}\Omega$ $R_2 = 10\text{K}\Omega$ $V_i = 300\text{mV (p-p)}$ frequency = 1KHz

$$V_o = 3.1\text{V (p-p)}$$

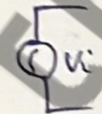
$$A = V_o/V_i = \frac{3.1\text{V}}{300\text{mV}} = 10.33 \rightarrow \text{Voltage amplification } (A_v) = 10.33$$

$$\text{For } V_i = 200\text{mV (p-p)}, V_o = 2\text{V (p-p)}$$

$$A = V_o/V_i = \frac{2\text{V}}{200\text{mV}} = 10 \rightarrow A_v = 10$$

$$\text{For } V_i = 100\text{mV (p-p)}, V_o = 1\text{V (p-p)}$$

$$A = V_o/V_i = \frac{1\text{V}}{100\text{mV}} = 10 \rightarrow A_v = 10$$

③ Now changing circuit from  to 

① (i) $R_1 = 100\text{K}\Omega$ $R_2 = 10\text{K}\Omega$ $V_i = 200\text{mV (p-p)}$ frequency = 1KHz

$$V_o = 1\text{V (p-p)}$$

$$A = \frac{V_o}{V_i} = \frac{1\text{V}}{200\text{mV}} = 5 \rightarrow \text{Voltage amplification } A_v = 5$$

(ii) $R_1 = 10\text{K}\Omega$ $R_2 = 1\text{K}\Omega$ $V_i = 200\text{mV (p-p)}$ frequency = 1KHz

$$V_o = 1.04\text{ (p-p)}$$

$$A = \frac{V_o}{V_i} = \frac{1.04}{200\text{mV}} = 5.2 \rightarrow \text{Voltage Amplification } (A_v) = 5.2$$

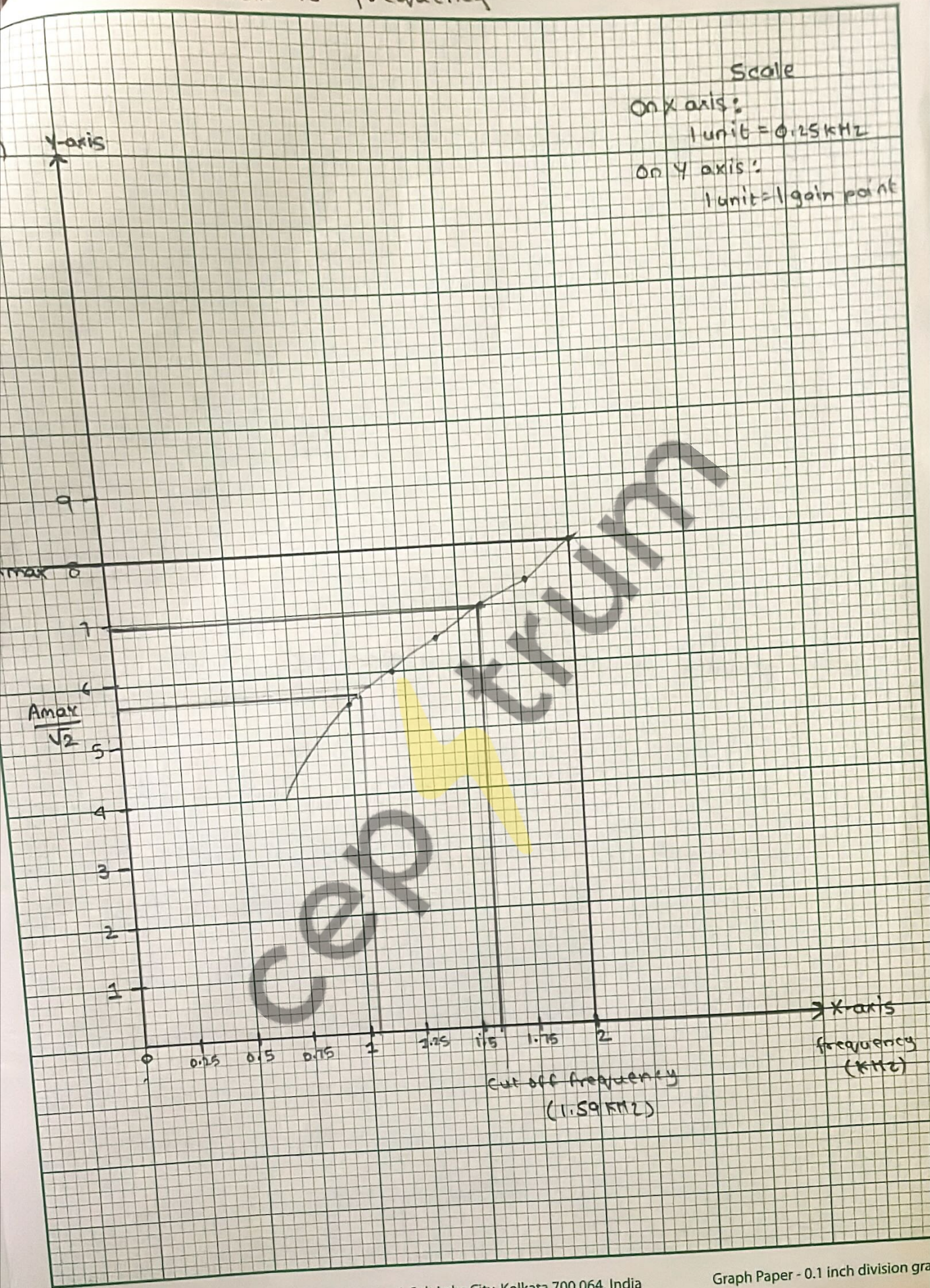
point: Non-Inverting High Pass Filter

$10\text{K}\Omega$ $R_F = 100\text{K}\Omega$ $R = 1\text{K}\Omega$ $C = 0.1\mu\text{F}$ $V_i = 200\text{mV (p-p)}$

cutoff frequency = 1.59KHz

frequency	V_o (output voltage)	Gain (V_o/V_i) = A
1KHz	1.1	5.5
1.1KHz		
1.2KHz	1.2	6
1.3KHz		
1.4KHz	1.3	6.5
1.5KHz		
1.6KHz (1.59)	1.4	7
1.7KHz		
1.8KHz	1.48	7.4
1.9KHz		
2.0KHz	1.6	8

Gain vs frequency



5) For a source with high internal impedance, a non-inverting op-amp will be suitable

Reason: For an inverting amplifier

$$A = - \frac{R_f}{R_i}$$

Here, A = gain;

R_f = output impedance

R_i = Input impedance

For a non-inverting amplifier $A = 1 + \frac{R_f}{R_i(NI)}$

Now, in comparing the magnitude of gain, by keeping A and R_f same in both cases

we see

$$R_i(NI) > R_i(I)$$

So we need non-inverting amplifier for high Internal impedance.